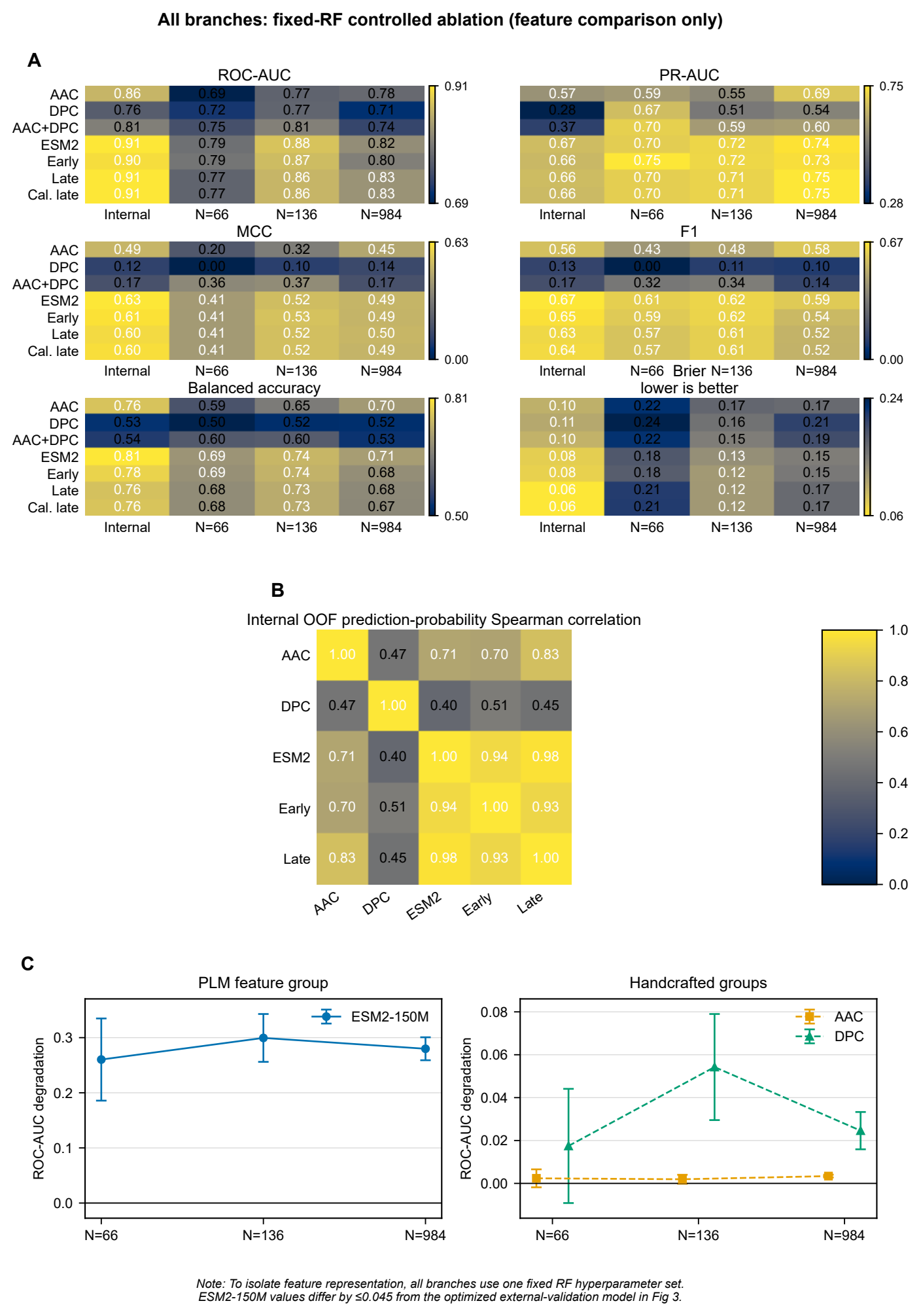


Note: To isolate feature representation, all branches use one fixed RF hyperparameter set.
ESM2-150M values differ by ≤0.045 from the optimized external-validation model in Fig 3.